table. However, following the in-context learning with demonstrations, this prevalence significantly diminishes. Meanwhile, there is a substantial increase in the proportion of cases involving begging the question. Our detailed exploration of reasoning outcomes reveals that, with demonstrations incorporated into the prompt, models acquire the ability to reason without resorting to shortcuts, thereby reducing the prevalence of questionable cause cases. Despite this improvement, models still struggle to consistently produce sound reasoning, often relying on generating reasoning from conclusions, as highlighted in the table. Similar to questionable cause, circular reasoning problems are solved in the few-shot setting, demonstrating that few-shot prompting prevents models from using shortcuts to reason.

C POV Reasoning

We further experimented with POV reasoning on a subset of results from the knowledge-enhanced experiments of the WinoBias dataset. The subset consists of 15 good examples, which are cases with perfect reasoning with correct answers, and the 45 bad examples, which are cases with less-than-perfect reasoning with still correct answers. We seek to use POV prompting to gain insights into internal reasoning process of the model and, in some cases, debug the reasoning. The prompt template for POV reasoning is shown in Table 13. The prompt template for POV diagnosis is shown in Table 11.

The results are summarized in Table 12, and one noticeable result is that the POV prompting shows similar accuracy for both good and bad reasoning samples. This possibly suggests that the model may not remain self-consistent beyond one-turn good reasoning. Conversely, POV reasoning enhances the model's ability to reason and self-correct in bad reasoning cases. Since LLMs are known to be sensitive to a given context, we break down the results by the correctness of the first turn prompt as shown in Table 12. The results show that the correctness of the first turn slightly increases accuracy for good reasoning cases and significantly increases accuracy for bad reasoning cases. This seems to mirror "first impression bias," in which people make quick and incomplete observations based largely on the first piece of information we receive.

In the case of good reasoning, introducing the incorrect first POV introduces higher Type3 SELF-CONTRA rate in particular, suggesting that the model might become self-contradictory in an effort to close the gap between the correct prediction and reasoning. In the case of bad reasoning, the incorrect first POV results in higher Type2 and Type3 SELF-CONTRA rates and significantly lower accuracy, which indicates that bad first information can easily lead the model off the track when the model inferences suspect reasoning. For future analysis, POV reasoning can be used for analyzing finer-grained categories of SELF-CONTRA errors.

C.1 POV as Diagnostic Tool

Moreover, we use POV as a diagnostic tool to gauge how confident LLMs are on their own reasoning. We ask the model to consider the pronoun in question from two points of view, and then ask the model to consider reasoning results and score how consistent the reasoning is on a scale of 1 to 10 with 10 being perfect.

We use POV reasoning as a diagnostic tool, as a way to perturb the reasoning and see how the model reacts. Then, we can gauge how confident the model feels about a particular line of reasoning. The results are summarized in Table 14. In general, we see that the model rates its reasoning to be rather consistent, though we see similar effects of "first impression bias" as before in Table 12. We see that wrong reasoning with Questionable Cause (QC) can be easily perturbed, which may suggest that the model is less confident about its reasoning in these cases. In contrast, the model is rather confident about Begging the Question (BQ) reasonings. We think this is reflective of the fallacy itself, where the model pre-concludes a reasoning path, and therefore, is very certain of the (wrong) foregone conclusion. Interestingly, the model scores the reasonings slightly lower for correct reasoning categories. Possibly, the model is considering many factors and paths in reasoning during inference rather than taking short-cuts or pre-supposed conclusions.

D Automatic evaluation

Besides the methods discussed in the main paper, we also experimented with other models, a vanilla Entailer (Tafjord et al., 2022) and a fine-tuned Flan-T5 (Chung et al., 2022) on our annotated examples, but both models performed rather poorly with precision of less than 0.1.

The prompts for automatic detection are shown

Categories	Results
Good reasoning	**Original question:** The janitor reprimanded [the accountant] because [he] made a mistake filing paperwork.
	First Turn: **Prompt:** The janitor reprimanded the accountant because he made a mistake filing paperwork. If we consider that "he" refers to the accountant, does the sentence make sense? What is your reasoning? **Model Response:** Yes, the sentence makes sense if we consider that "he" refers to the accountant. The sentence states that the janitor reprimanded the accountant because he (the accountant) made a mistake filing paperwork. The use of "he" as a pronoun in this context is clear and logical, as it refers back to the previously mentioned noun "accountant".
	Second Turn: **Prompt:** This time, consider that the janitor is "he". Does the sentence make sense? What is your reasoning? **Model Response:** If we consider that the janitor is "he", the sentence does not make sense. The sentence states that the janitor reprimanded the accountant because he (the janitor) made a mistake filing paperwork. However, it is not logical for the janitor to reprimand the accountant for his own mistake. In this case, the sentence would need to be rephrased to clarify the intended meaning.
	Final Turn: **Prompt:** Based on your previous considerations for both points of views, who is "he" more likely to be and why? **Model Response:** Based on the previous considerations, 'he' is more likely to refer to the accountant. This is because the sentence states that the janitor reprimanded 'him' (the accountant) for making a mistake filing paperwork. The use of 'he' as a pronoun in this context is consistent with the subject of the sentence, which is the accountant. Therefore, it is more logical to interpret 'he' as referring to the accountant rather than the janitor.
Bad reasoning	**Original question:** [The analyst] loved to get a haircut from the hairdresser because [she] could sit on a comfy chair.
	First Turn: **Prompt:** The analyst loved to get a haircut from the hairdresser because she could sit on a comfy chair. If we consider that "she" refers to the hairdresser, does the sentence make sense? What is your reasoning? **Model Response:** No, the sentence does not make sense if we consider that "she" refers to the hairdresser. The reason is that the sentence states that the analyst loved to get a haircut from the hairdresser because "she" could sit on a comfy chair. However, it is the analyst who is getting the haircut, not the hairdresser. Therefore, if "she" refers to the hairdresser, the sentence becomes illogical.
	Second Turn: **Prompt:** This time, consider that the analyst is "she". Does the sentence make sense? What is your reasoning? **Model Response:** Yes, if we consider that the analyst is 'she', the sentence makes sense. The sentence states that the analyst loved to get a haircut from the hairdresser because "she" could sit on a comfy chair. In this context, "she" refers to the analyst, indicating that the analyst enjoyed the experience of sitting on a comfy chair while getting a haircut from the hairdresser. The sentence is logical and coherent when interpreted in this way.
	Final Turn: **Prompt:** Based on your previous considerations for both points of views, who is "she" more likely to be and why? **Model Response:** Based on the previous considerations, "she" is more likely to refer to the hairdresser rather than the analyst. The main reason is that the sentence states that "she" could sit on a comfy chair, which implies that the person referred to as "she" is the one receiving the haircut. Since the analyst is the one getting the haircut, it would be more logical for "she" to refer to the hairdresser who is providing the service. Therefore, the hairdresser is more likely to be "she" in this sentence.

Table 9: **Examples of POV**

Method	Correct Reasoning		Wrong Reasoning		
	Evidence Missing	Incomplete Reasoning	Questionable Cause	Begging the Question	Wrong Context Knowledge
Zero-shot (R)	0.75 (20) / 0.09	0.25 (20) / 0.03	0.83 (35) / 0.33	0.0 (35) / 0.035	0.06 (35) / 0.02
Zero-shot (A)	0.571 (7) / 0.03	0.429 (7) / 0.015	0.60 (60) / 0.465	0.117 (60) / 0.08	0.2 (60) / 0.07
Few-shot (R)	0.614 (44) / 0.255	0.386 (44) / 0.125	0.20 (30) / 0.03	0.70 (30) / 0.195	0.10 (30) / 0.065
Few-shot (A)	0.50 (24) / 0.115	0.50 (24) / 0.08	0.121 (33) / 0.055	0.906 (33) / 0.285	0.0 (33) / 0.0

Table 10: **Results of Finer-grained categories** For each result, we provide dual perspectives by reporting the proportions of case counts relative to both Type1 and Type2 cases, as well as the entire dataset consisting of 200 datapoints. In correct reasoning, the initial number is derived from Type1 cases, while in wrong reasoning, the initial number is based on Type2 cases. The total numbers for Type1 and Type2 cases are indicated in parentheses. For example, 0.75 (20) means there are 20 Type1 cases in zero-shot (R) and 15 of them are evidence missing categories. We highlight questionable cause results in zero-shot setting and begging the question in few-shot setting because those two have the highest ratios.

in Table 15 and Table 16. The human detection guideline is shown in Figure 5, 6, 7 and the demonstrations we show to humans are the same as the model. Due to the length of the demonstrations, we will release demonstrations in the code upon publication. We calculate the final results using the model's output. For the FGA, we first give the model the definition and examples of finer-grained categories as shown in the guideline and then ask the model if the reasoning has such problems (e.g., questionable cause). For evidence missing, we will separate the input question into multiple evidence segments and ask the model if the reasoning mentions those evidence segments one by one.

You will be given questions, their corresponding answer-reasoning pairs and gold labels. Your task is to annotate: self-contra rate (binary label), finer-grained. If you have any notes, you can leave notes in the column after (optional).
You need to annotate finer-grained categories for every instance. If the reasoning does not have a problem, you can leave it blank.

Self-contra: reasoning itself is self-contradictory. For example, the question is “The cleaner got mad at the farmer because she brought lots of dirt. Who is “she"?”, and the reasoning is “The cleaner got mad at the farmer because the farmer brought lots of dirt. Since the cleaner is the one who got mad, it can be inferred that “she" refers to the cleaner.” The reasoning itself already points out the answer is the farmer but claims the opposite answer later.

Your data will be used in the research project for self-contradictory reasoning study.

Figure 5: Introduction of task for human detection.

Wrong reasoning types:

1. Questionable cause also known as causal fallacy, means a cause is incorrectly identified. In our case, it usually means models use shortcuts or syntax to conclude instead of using evidence and semantic meaning, such as “since the carpenter is the subject”, “since the carpenter is mentioned earlier”, etc.
   Add one example: how model using shortcuts

2. Begging the question is a logical fallacy where the conclusion of an argument is assumed in one of the premises, essentially assuming the truth of what one is trying to prove. It occurs when the argument's premises already presuppose the truth of the conclusion, making the argument circular and not providing any real evidence or support for the conclusion. [The conclusion is part of the premise, evidence]
   For example:
   The question is "The janitor sends the editor sorted newspapers every day because he required previously.who is 'he'".
   And the reasoning is: Since the janitor is the one who sends the newspapers, it can be inferred that 'he' refers to the janitor who required the task to be done previously.
   This one assumes the janitor is the answer first, then uses this answer to conclude, and the argument doesn't offer any independent or external evidence to support the conclusion. It's important to avoid begging the question in logical reasoning to ensure the validity of an argument.

3. Wrong context knowledge means that the reasoning captures the wrong information from the context, which is the question in our case. For example, there are two characters in the context, but models say there is only one character.

Figure 6: Wrong reasoning explanation for human detection.